Cambridge (CIE) IGCSE Computer Science

SQL

Your notes

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SQL

What is SQL?

- **SQL** (Structured Query Language) is a programming language used to interact with a **DBMS**.
- The use of SQL allows a user to:
  - **Select data**
  - **Order data**
  - **Sum data**
  - **Count data**

Selecting data commands

Command	Description	Example
SELECT	Retrieves data from a database table	**SELECT** * **FROM** users; (retrieves all data from the 'users' table) **SELECT** name, age **FROM** users (retrieves names and ages from the 'users' table)
FROM	Specifies the tables to retrieve data from	**SELECT** name, age **FROM** users; (retrieves names and ages from the 'users' table)
WHERE	Filters the data based on a specified condition	**SELECT** * **FROM** users **WHERE** age > 30; (Retrieves users older than 30)
AND	Combines multiple conditions in a **WHERE** clause	**SELECT** * **FROM** users **WHERE** age > 18 **AND** city = 'New York'; (retrieves users older than 18 and from New York)

OR	Retrieves data when at least one of the conditions is true	**SELECT** * **FROM** users **WHERE** age < 18 **OR** city = 'New York'; (retrieves users younger than 18 or from New York)
WILDCARDS	'*****' and '**%**' symbols are used for searching and matching data '*****' used to select all columns in a table '**%**' used as a wildcard character in the LIKE operator	**SELECT** * **FROM** users; (retrieves all columns for the 'users' table) **SELECT** * **FROM** users **WHERE** name **LIKE** 'J%'; (retrieves users whose names start with 'J')
ORDER BY	How data is organised (sorted) when it is retrieved	**SELECT** Forename, Lastname **FROM** Students **WHERE** StudentID < 10 **ORDER BY** Lastname, Forename **ASC** (retrieves only the *forename* and *lastname* of all students from the *students* table who have a *studentID* of less than 10 and displays in ascending order by *lastname* and *forename*)
SUM	Adds up and outputs the sum of a field	**SELECT SUM**(Salary) **FROM** tbl_people;
COUNT	Counts the number of files which match the set criteria	**SELECT COUNT**(*) **FROM** tbl_people **WHERE** Salary > 50000;

Examples

- Select all the fields from the Customers table

Command:

```
SELECT * FROM Customers;
```

Output:

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ID	Name	Age	City	Country
1	John Doe	30	New York	USA
2	Jane Doe	25	London	UK
3	Peter Lee	40	Paris	France

- Select the ID, name & age of customers who are older than 25

Command:

```
SELECT ID, name, age
FROM Customers
WHERE Age > 25;
```

Output:

ID	Name	Age
1	John Doe	30
3	Peter Lee	40

- Select the name and country of customers who are from a country that begins with 'U'

Command:

```
SELECT Name, Country
FROM Customers
WHERE Country LIKE 'U%';
```

Output:

Name	Country

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John Doe	USA
Jane Doe	UK

- Select all fields of customers who are from 'London' or 'Paris'

Command:

```
SELECT *
FROM Customers
WHERE City = 'London' OR City = 'Paris';
```

Output:

ID	Name	Age	City	Country
2	Jane Doe	25	London	UK
3	Peter Lee	40	Paris	France

Worked Example

Below is a table of animals called ***tbl_animals***

Animal	Breeding	Number of Young
Red Fox	Yes	4–6
Rabbit	Yes	4–12
African Elephant	Yes	1
Blue Whale	No	1
Orangutan	Yes	1
Polar Bear	Yes	1–3
Dolphin	Yes	1

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Kangaroo	Yes	1
Lion	Yes	1–6
Penguin	Yes	1

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Complete this SQL statement to display all of the animal breeds that are currently breeding and there was only one young born this year **[3]**

SELECT ....

FROM ....

WHERE ....

Answer

SELECT Animal [1]

FROM tbl_animals [1]

WHERE Breeding = "Yes" AND Number of Young = 1 [1]